

MAHMOUD ELLOUMI

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PROFILE

Computer Science and Applied Mathematics engineering student at ENIS. Across two internships and several projects, I have built full-stack applications powered by LLMs: conversational and data assistants running locally with Ollama, document extraction with Mistral AI, and automated workflows with n8n. Seeking a **final-year internship (PFE)** to design **AI agents** that solve real business needs.

EXPERIENCE

- Full-Stack & AI Developer Intern – IDSOFT (ID Software Solutions)**

Sfax | Jul. – Aug. 2026

AI-enhanced after-sales service (SAV) management app

React, Spring Boot, PostgreSQL, Ollama
- Web app covering the full after-sales lifecycle (7-state machine), with three portals: customer, technician, manager.
 - 6 local AI features (Ollama): troubleshooting assistant, natural-language data assistant, request classification, best-technician suggestion, report-writing assistance.
 - Spring Boot REST API (Swagger), dashboard; security with JWT, TOTP, BCrypt, role-based access control.
- Mobile Developer Intern – LOGICOM Informatique**

Sfax | Jul. – Aug. 2025

Product scanning mobile app

Flutter
- Flutter mobile app that scans a product’s QR code and displays its available prices and details.
 - Connected to the company database to retrieve prices and product records.

PROJECTS

- SPARKY – Intelligent Business Document Processing (PFA)**

May 2026

Team of 3, Scrum

Angular, FastAPI, Mistral AI, PostgreSQL, n8n
- Automated digitization, classification and data extraction for invoices, quotes, purchase orders and delivery notes, using Mistral AI-based OCR.
 - 3-tier architecture (Angular 20, asynchronous FastAPI REST API, PostgreSQL), supplier price comparison and analytics dashboard.
 - Workflows and notifications orchestrated with n8n; comparative study with a local n8n + Ollama (Qwen2.5) prototype.
- Intelligent Traffic Monitoring**

2025
- Real-time vehicle detection and classification on video streams with YOLO and OpenCV (Python).
 - Traffic analytics dashboard: vehicle counts and breakdown by vehicle type.
- TERRANAUT – Drone-Based Detection of Sick Animals**

2024
- Team project: detecting sick animals from thermal images captured by drones.
 - Applied computer vision and deep learning (object detection) to thermal image analysis.

TECHNICAL SKILLS

AI & LLMs	LLMs (Ollama, Mistral AI, Qwen), AI agents, Machine Learning, Deep Learning
Vision & Automation	OpenCV, YOLO, PyTorch, n8n
Languages	Python, Java, JavaScript/TypeScript, C
Web & Mobile	React, Angular, Spring Boot, FastAPI, Node.js, Flutter
Data & DevOps	PostgreSQL, MySQL, MongoDB, Kafka, Docker, Git, Linux, Scrum

EDUCATION

Engineering Degree in Computer Science and Applied Mathematics – ENIS, Sfax	2024 – Present
Preparatory Cycle in Technology – IPEIS, Sfax (<i>top 100 in the national ranking</i>)	2022 – 2024
Baccalaureate in Technical Sciences – 15 November 1955 High School, Sfax	2022

CERTIFICATIONS

NVIDIA Getting Started with Deep Learning • Cisco C Advanced • Scrum (SFC) • Hedera (HCF) • Microsoft MS-900

VOLUNTEERING (IEEE)

Photographer – IEEE ENIS SB (2026) • Treasurer & Webmaster – IEEE Cyber Security ENIS (2025)
Social Media Manager & Photographer – TSYP 13 • Organizer – Pitch Yourself for a Career

LANGUAGES & INTERESTS

Arabic: native French: upper-intermediate English: intermediate Spanish: elementary
Interests: photography, art, sports, innovation